

REU-HPC: Week-3 Group Meeting

Group Meetings

Tuesday 3-4pm (starting June 16)

<https://utexas.zoom.us/j/85978334209>

Week 3 Objectives (June 15 – June 21)

The primary goal for week 2 and 3 is to become familiar with:

- PEAK profiler and software/package installation
Get familiar with the PEAK profiler, including its basic workflow and usage. Also learn how to install software packages and dependencies on TACC systems.
- The project abstract is due on June 18.

Useful information

- **Frontera-top100-2025-1.txt:**

This file contains the top 100 executables on Frontera from January to June 2025, sorted by core-hours. Only executable names are provided. We need to identify the corresponding software package (if there's one).

Link: <https://drive.google.com/file/d/15VXVPnREFcz3hiVyVoCW8ThX5TO6WELO/view?usp=sharing>

- **package_list.xlsx**

This spreadsheet summarizes the software packages that we are currently interested in studying.

Link:

<https://docs.google.com/spreadsheets/d/1JSHFZpOV8IA3M1mw86WJAuWZ8kotStPf/edit?usp=sharing&oid=101096215360805228479&rtpof=true&sd=true>

- Project directory: **/work2/05392/cylu/share/reu_2026/2.project**

The Directory **1.build_scripts** contains the build scripts for PEAK and several scientific packages. There are two subdirectories:

1.peak: **build_peak.sh** is the basic build script that builds PEAK using the default configuration.

2.apps: This directory contains build scripts for selected scientific applications. Currently available build scripts include: *lammps*, *yambo*, *berkeley_gw*, *gromacs*, *dftb+*, *abinit*, *meep*. More build scripts will be added as the project progresses.

Things to Try

1. **Build available packages on Stampede3**
Start with the packages for which build scripts are already available and try building them on Stampede3.
2. **Run simple test cases**
Run simple test cases for each package. Some sample cases will be added to the project directory. You may also look for sample or tutorial cases from the official websites of the corresponding packages.
3. **Run the package with PEAK**
Use PEAK to profile the package. Set the following environment variable to check whether the package uses BLAS, LAPACK, or FFTW functions:

```
export PEAK_TARGET_GROUP=blas,lapack,fftw
```

Then run the application with PEAK and examine the profiling report to identify any BLAS, LAPACK, or FFTW function calls.